

Last-Mile Delivery Tracker

Objective:

- Logistics operations involve complex pricing rules, dynamic agent assignment, and reliable customer communication. Build a delivery management platform where customers and admins can create orders with auto-calculated charges, agents are assigned intelligently, and customers are notified at every step of the delivery journey.

Scope of Work:

- **Input:** Pickup & drop address, package dimensions (L×B×H), actual weight, order type (B2B/B2C), payment type (Prepaid/COD)
- **Output:** Order with auto-calculated charge, agent assignment, status tracking, notifications
- Admin manages zones, assigns areas to zones, and configures rate cards (intra and inter-zone rates separately for B2B and B2C) and COD surcharge per order type
- Customer can register, log in, and place an order; Admin can also create orders on behalf of a customer
- On order creation, system detects pickup and drop zones, calculates volumetric weight ($L \times B \times H \div 5000$), bills on higher of actual vs volumetric weight, applies zone rate from correct rate card (B2B or B2C), and adds COD surcharge if applicable. The charge is shown before the customer confirms
- Admin can manually assign a delivery agent to an order or trigger auto-assignment to the nearest available agent
- Delivery agent updates order status (Picked Up / In Transit / Out for Delivery / Delivered / Failed)
- On failed delivery, customer receives notification and can reschedule for a new date; agent is reassigned for the rescheduled attempt
- Customer can view live order status and full tracking timeline
- Email notifications sent to customer on every status change
- Admin can view all orders, filter by status/zone/agent, and override any order status

Technical Expectations:

- Backend API, Frontend, Database with role-based auth (customer / delivery agent / admin)
- Rate calculation engine: zone detection, volumetric weight, B2B/B2C rate card lookup, COD surcharge, all admin-configurable, no hardcoding
- Auto-assignment logic: detect nearest available agent based on current location or zone
- Order status lifecycle with immutable tracking history (each status change logged with timestamp and actor)
- Failed delivery flow: status flagged, customer notified, reschedule captured, agent reassigned
- Email and SMS integration for status notifications (any free tier service)

Deliverables:

1. Zip file with complete source code
2. README with setup guide, .env.example, API docs, DB schema, and rate calculation logic explanation
3. Hosted application URL (Vercel, Render, Railway, or similar)
4. System design write-up (800 words max) covering rate calculation engine, zone detection approach, auto-assignment logic, and failed delivery handling

Evaluation Focus:

- Rate calculation engine design and correctness (zone, volumetric weight, B2B/B2C, COD)
- Auto-assignment logic and agent availability modelling
- Order status lifecycle and immutable tracking history
- Database schema and data modelling
- API design and code structure
- Documentation